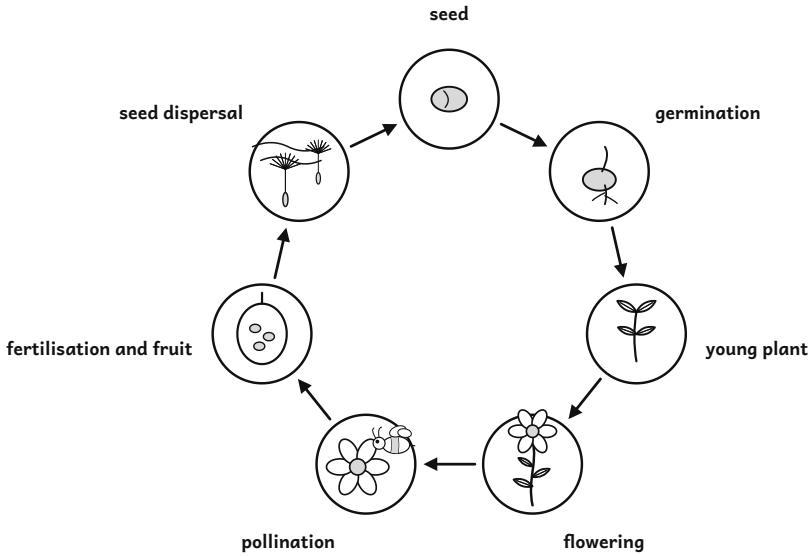
LIVING THINGS

Plant Growth and Development

Flowering and non-flowering plants, the life cycle of a flowering plant and measuring growth



In this pack

📖 **Learn it:** How plants grow ★☆☆
Fact check: Plant life cycle ★☆☆
Think it through: Growing and comparing ★★☆
Investigate: How fast does a bean plant grow? ★★☆
Real-world case study: A rice farmer's year ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · A2 · PLANT GROWTH

How plants grow

★☆☆ Grade 3

Name: Class: Date:

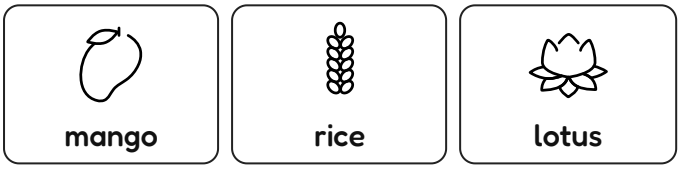
Learning goals

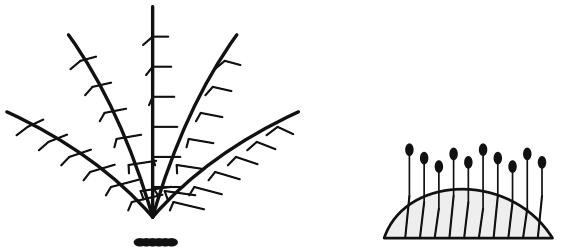
- Tell flowering and non-flowering plants apart.
- Describe the life cycle of a flowering plant and its main parts.
- Measure plant growth and compare plant and animal life cycles.

A tiny seed can grow into a giant mango tree. How does it happen? Let's follow a plant through its life.

Flowering and non-flowering plants

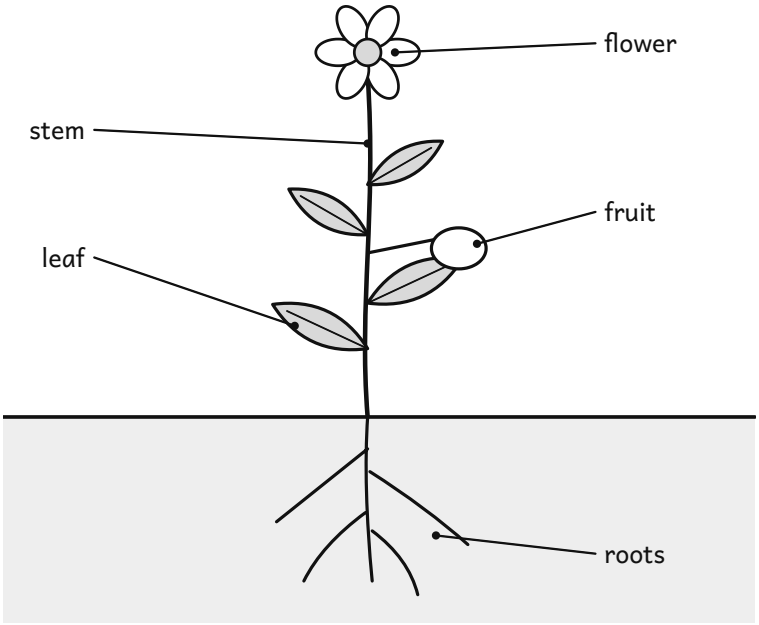
Flowering plants grow flowers that make seeds: mango, rice, lotus, sunflower, grass. **Non-flowering plants** do not have flowers: **ferns** and **mosses** grow from tiny **spores**, and pine trees make seeds in **cones**.





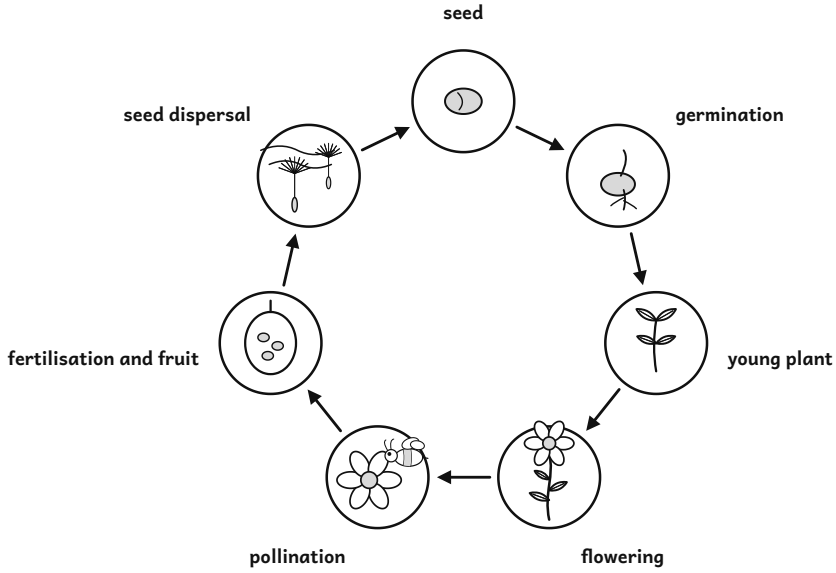
Ferns and mosses: no flowers.

The parts of a flowering plant



Root, stem, leaf, flower and fruit.

The life cycle of a flowering plant



From seed to seed.

- A **seed germinates** (starts to grow) when it has water and warmth.
- It becomes a **seedling** with its first leaves.
- The seedling grows into a **mature** (adult) plant.
- The plant makes **flowers**. Insects or wind carry pollen, and **seeds** form inside **fruits**.
- The seeds are spread, and the cycle starts again.

Measuring growth

We can measure a plant's **height** with a ruler each week, count its **leaves**, or measure the width of its stem. A bar chart shows how it grows over time.

Plants and animals

	Flowering plant	Chicken
starts as	seed	egg
young	seedling	chick
adult	plant with flowers	hen
new life from	seeds	eggs

Bangladesh connection

Rice is the most important crop in Bangladesh. Farmers plant rice seedlings in flooded fields and harvest the grains (seeds) a few months later.

Key words

germinate start to grow from a seed
seedling a young plant
mature fully grown
spore a tiny part that grows into a fern or moss
life cycle the stages of life, from seed to seed

FACT CHECK · A2 · PLANT GROWTH

Plant life cycle

★☆☆
Grade 3

Name: Class: Date:

Learning goals

- Sort plants.
- Order the life cycle.

1 Sort the plants. [6]

SORT THESE

rice **mango tree** **fern** **pine tree** **sunflower** **moss**

Flowering	Non-flowering

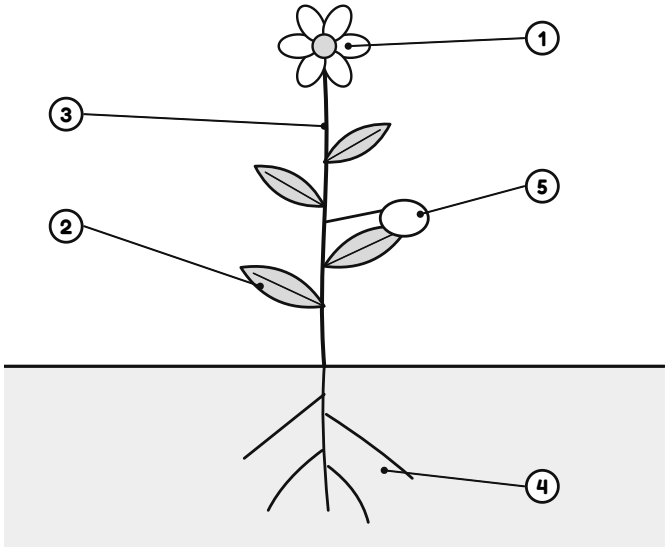
2 Number the stages in order. [4]

- ☐ fruit with new seeds
- ☐ seed
- ☐ adult plant with flowers
- ☐ seed germinates
- ☐ seedling

3 **Label the plant.** [5]

LABELS

flower leaf roots stem fruit



1 _____ 2 _____ 3 _____

4 _____ 5 _____

THINK IT THROUGH · A2 · PLANT GROWTH

Growing and comparing

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Compare life cycles.
- Explain plant needs.

1 **How is *a* plant life cycle like *a* chicken's? How is it different?** [2]

2 **A seed is kept dry in *a* packet. Will it germinate? Why?** [2]

3 **Draw the life cycle of *a* bean plant with four stages and arrows.** [3]

Bean life cycle

INVESTIGATE · A2 · PLANT GROWTH

How fast does a bean plant grow?

★★☆ Grade 3

Name: **Class:** **Date:**

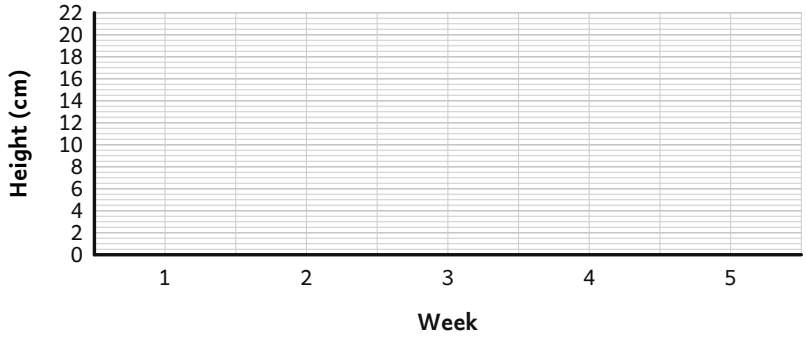
Learning goals

- Measure height each week.
- Draw a bar chart.
- Find a pattern.

Class 3C measured a bean plant every week.

1 **Draw a bar chart.** [3]

Week	1	2	3	4	5
Height (cm)	2	6	11	17	20



2 **Describe the pattern.** [1]

3 **In which week did it grow the most?** [1]

4 **Predict the height in week 6.** [1]

REAL-WORLD CASE STUDY · A2 · PLANT GROWTH

A rice farmer's year

★★★ Grade 3

Name: Class: Date:

Learning goals

- Follow a real crop through its life cycle.

Science | Farming | Maths

Abdul is a rice farmer in Mymensingh. First he soaks rice seeds in water until they **germinate**. He plants them close together in a small bed of mud. After about a month the **seedlings** are moved to the big flooded field.

The rice plants grow tall and green. Tiny **flowers** appear, and the wind carries pollen. The grains (**seeds**) grow and turn golden. About four months after planting, Abdul **harvests** the rice. He keeps some seeds to plant next year.

Based on rice farming in Bangladesh.

1 **Put Abdul's year in order.** [4]

- ☐ harvest the grains
- ☐ soak seeds so they germinate
- ☐ rice flowers
- ☐ grow seedlings in a mud bed
- ☐ move seedlings to the field

2 **Why does Abdul keep some seeds?** [1]

.....

3 **Rice takes about 4 months from planting to harvest. How many weeks is that?** [1]

.....

REVIEW AND CHALLENGE · A2 · PLANT GROWTH

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use plant words.

1 Find the words.

C	P	N	E	G	R	O	W	L
R	N	F	E	R	N	K	B	P
W	O	N	A	R	R	O	O	T
J	M	N	F	L	O	W	E	R
C	S	G	S	T	E	M	O	R
F	E	L	K	S	S	P	S	V
C	E	E	I	W	E	E	K	I
V	D	A	K	V	Y	H	L	O
I	B	F	T	F	R	U	I	T

fern flower fruit leaf root seed stem

2 Which plant has no flowers? [1]

- ◯ mango
- ◯ fern
- ◯ rice

3 A seed starting to grow is called... [1]

- ◯ germinating
- ◯ flowering
- ◯ wilting

4 Challenge. Why do plants make so many seeds? [2]

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
sort flowering and non-flowering plants	○	○	○
name the parts of a plant	○	○	○
draw a plant life cycle	○	○	○
measure plant growth	○	○	○
compare plant and animal life cycles	○	○	○